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Light emitting diode devices with bonding and/or ohmic contact-reflective material

Abstract

A metal stack of layers contacting an N-type layer of a light emitting diode (LED) device comprises: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer; a reflective layer electrically contacting the ohmic contact layer; a first material barrier layer electrically contacting the reflective layer; a current carrying layer electrically contacting the first material barrier layer; and a second material barrier layer electrically contacting the current carrying layer. LED devices incorporate the metal stack of layer as a bonding material and/or as an ohmic contact-reflective material. Methods of making and using the metal stacks and LED devices are also provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Application No. 63/240,589, filed Sep. 3, 2021, the entire disclosure of which is hereby incorporated by reference herein.

TECHNICAL FIELD

(1) Embodiments of the disclosure generally relate to light emitting diode (LED) devices and methods of manufacturing the same. More particularly, embodiments are directed to metal stacks and LED devices incorporating the same. The metal stacks are in contact with an N-type layer of the devices. The metal stacks comprise: an ohmic contact layer, a reflective layer, a first material barrier layer, a current carrying layer, and a second material barrier layer.

BACKGROUND

(2) A light emitting diode (LED) is a semiconductor light source that emits visible light when current flows through it. LEDs combine a p-type semiconductor with an n-type semiconductor. LEDs commonly use a III-V group compound semiconductor. A III-V group compound semiconductor provides stable operation at a higher temperature than devices that use other semiconductors. The III-V group compound is typically formed on a substrate formed of sapphire aluminum oxide (Al₂O₃) or silicon carbide (SiC).

(3) High-intensity/brightness light emitting devices capable of operation across the visible spectrum include Group III-V semiconductors, particularly binary, ternary, and quaternary alloys of gallium, aluminum, indium, and nitrogen, also referred to as III-nitride materials. Typically, III-nitride light emitting devices are fabricated by epitaxially growing a stack of semiconductor layers of different compositions and dopant concentrations on a growth substrate such as a sapphire, silicon carbide, III-nitride, or other suitable substrate by metal-organic chemical vapor deposition (MOCVD), molecular beam epitaxy (MBE), or other epitaxial techniques. Sapphire is often used as the growth substrate due to its wide commercial availability and relative ease of use. The stack grown on the growth substrate typically includes one or more n-type layers doped with, for example, Si, formed over the substrate, a light emitting or active region formed over the n-type layer or layers, and one or more p-type layers doped with, for example, Mg, formed over the active region.

(4) Low work function metals are usually employed to form ohmic contact to n-GaN. Currently, the state-of-the-art ohmic contact materials for n-GaN are aluminum (Al) and titanium (Ti) metals, and low contact resistance to n-GaN is achieved, attributing to low work function of the Al and Ti materials. Due to its better optical reflectivity and electrical conductivity of Al as compared to Ti, Al is commonly used as n-GaN ohmic contact metal for high power LED devices. Al optical reflectivity is much lower than reflective materials such as silver (Ag) and gold (Au), especially at low optical angle of incident, and light long wavelength. Ag or Au, however, is not a good ohmic contact material to n-GaN, due to its high material work function, resulting extreme high contact resistance to n-GaN.

(5) There is a need in the art to provide metal stacks that are suitable for providing ohmic contact and reflectivity to N-type layers.

SUMMARY

(6) Provided herein are light emitting diode (LED) devices and methods of making and using the same.

(7) In an aspect, a light emitting diode (LED) device comprises: semiconductor layers including an N-type layer, an active region, and a P-type layer; a metal stack of layers contacting the N-type

layer, and a dielectric material which insulates the P-type layer and the active region from the N-bonding material. The metal stack of layers comprises: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, a reflective layer electrically contacting the ohmic contact layer, a first material barrier layer electrically contacting the reflective layer, a current carrying layer electrically contacting the first material barrier layer, and a second material barrier layer electrically contacting the current carrying layer.

(8) A further aspect is a method of manufacturing a metal stack of a light emitting diode (LED) device, the method comprising: depositing an ohmic contact layer electrically contacting an N-type layer of the LED device and having a work function value that is less than or equal to a work function value of the N-type layer; depositing a reflective layer electrically contacting the ohmic contact layer; depositing a first material barrier layer electrically contacting the reflective layer; depositing a current carrying layer electrically contacting the first material barrier layer; and depositing a second material barrier layer electrically contacting the current carrying layer.

(9) In another aspect, a light emitting diode (LED) device comprises: semiconductor layers including an N-type layer, an active region, and a P-type layer; an N-bonding material electrically contacting the N-type layer; a P-bonding material electrically contacting the P-type layer and being isolated from the N-bonding material; and a first dielectric material which insulates the P-type layer and the active region from the N-bonding material. The N-bonding material being multilayered and comprising: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, a reflective layer electrically contacting the ohmic contact layer, a first material barrier layer electrically contacting the reflective layer, a current carrying layer electrically contacting the first material barrier layer, and a second material barrier layer electrically contacting the current carrying layer.

In one or more embodiments, the LED device is in the form of a chip scale package. In one or more embodiments, the LED device is in the form of a micro-LED monolithic array.

(10) A further aspect is a method of manufacturing a light emitting diode (LED) device, the method comprising: preparing a plurality of semiconductor layers including an N-type layer, an active region, and a P-type layer; exposing the N-type layer; preparing a bonding material contacting the N-type layer; and depositing a dielectric material which insulates the P-type layer and the active region from the N-bonding material. The bonding material is formed by: depositing an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer; depositing a reflective layer electrically contacting the ohmic contact layer; depositing a first N-bonding material barrier layer electrically contacting the reflective layer; depositing a current carrying layer electrically contacting the first material barrier layer; and depositing a second N-bonding material barrier layer electrically contacting the current carrying layer.

(11) In an aspect, a light emitting diode (LED) device comprises: a plurality of mesas, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each mesa having a top surface and at least one mesa sidewall; a plurality of trenches between the mesas defined by respective mesa sidewalls and each having a bottom surface, each of the trenches containing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas; an N-electrode metal contained by the N-ohmic contact-reflective material, a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material; and a P-electrode metal in electrical contact with the P-type layer of each of the mesas. The N-ohmic contact-reflective material is multilayered and comprises: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, a reflective layer electrically contacting the ohmic contact layer, a first material barrier layer electrically contacting the reflective layer, a current carrying layer electrically contacting the first

material barrier layer, and a second material barrier layer electrically contacting the current carrying layer.

(12) A further aspect is a method of manufacturing a light emitting diode (LED) device, the method comprising: preparing a plurality of mesas and trenches, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each mesa having a top surface and at least one mesa sidewall, and the trenches defined by respective mesa sidewalls and each having a bottom surface; exposing the N-type layer; preparing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas; depositing and patterning an N-electrode metal contained by the N-ohmic contact-reflective material and a P-electrode metal in electrical contact with the P-type layer of each of the mesas; and depositing and patterning a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material. The N-ohmic contact-reflective material being prepared by: depositing an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, depositing a reflective layer electrically contacting the ohmic contact layer, depositing a first material barrier layer electrically contacting the reflective layer, depositing a current carrying layer electrically contacting the first material barrier layer, and depositing a second material barrier layer electrically contacting the current carrying layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments. The embodiments as described herein are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements. The figures herein are not to scale.

(2) FIG. 1 a cross-sectional view of a metal stack of layers according to one or more embodiments;

(3) FIG. 2 illustrates a cross-sectional view of an LED device, in particular, a chip scale package, according to one or more embodiments;

(4) FIG. 3 illustrates a cross-sectional view of an LED device, in particular, a monolithic uLED array, according to one or more embodiments;

(5) FIG. 4A illustrates a cross-sectional view of an LED device, in particular, an array of uLEDs on a substrate, according to one or more embodiments, and FIG. 4B illustrates the cross-sectional view of the LED device of FIG. 4A prior to depositing material the trenches;

(6) FIG. 5 is a transmission electron microscope (TEM) micrograph of an excerpt of an exemplary embodiment made according to methods herein;

(7) FIG. 6 provides an exemplary process flow diagram for manufacture of a metal stack according to one or more embodiments;

(8) FIG. 7 provides an exemplary process flow diagram for manufacture of a LED device according to one or more embodiments;

(9) FIG. 8 schematically illustrates a top view of an exemplary display device according to one or more embodiments; and

(10) FIG. 9 schematically illustrates an exemplary display system comprising LED devices according to embodiments herein.

DETAILED DESCRIPTION

(11) Before describing several exemplary embodiments of the disclosure, it is to be understood that the disclosure is not limited to the details of construction or process steps set forth in the following description. The disclosure is capable of other embodiments and of being practiced or being carried out in various ways.

(12) The term “substrate” as used herein according to one or more embodiments refers to a structure, intermediate or final, having a surface, or portion of a surface, upon which a process acts. In addition, reference to a substrate in some embodiments also refers to only a portion of the substrate, unless the context clearly indicates otherwise. Further, reference to depositing on a substrate according to some embodiments includes depositing on a bare substrate, or on a substrate with one or more films or features or materials deposited or formed thereon.

(13) In one or more embodiments, the “substrate” means any substrate or material surface formed on a substrate upon which film processing is performed during a fabrication process. In exemplary embodiments, a substrate surface on which processing is performed includes materials such as silicon, silicon oxide, silicon on insulator (SOI), strained silicon, amorphous silicon, doped silicon, carbon doped silicon oxides, germanium, gallium arsenide, glass, sapphire, and any other suitable materials such as metals, metal nitrides, III-nitrides (e.g., GaN, AlN, InN and alloys), metal alloys, and other conductive materials, depending on the application. Substrates include, without limitation, light emitting diode (LED) devices, including uLED devices. Substrates in some embodiments are exposed to a pretreatment process to polish, etch, reduce, oxidize, hydroxylate, anneal, UV cure, e-beam cure and/or bake the substrate surface. In addition to film processing directly on the surface of the substrate itself, in some embodiments, any of the film processing steps disclosed are also performed on an underlayer formed on the substrate, and the term “substrate surface” is intended to include such underlayer as the context indicates. Thus for example, where a film/layer or partial film/layer has been deposited onto a substrate surface, the exposed surface of the newly deposited film/layer becomes the substrate surface.

(14) The term “wafer” and “substrate” will be used interchangeably in the instant disclosure. Thus, as used herein, a wafer serves as the substrate for the formation of the LED devices described herein.

(15) Reference to a micro-LED (uLED or μ LED) means a light emitting diode having one or more characteristic dimensions (e.g., height, width, depth, thickness, etc. dimensions) of less than 100 micrometers. In one or more embodiments, one or more dimensions of height, width, depth, thickness have values in a range of 2 to 25 micrometers.

(16) Advantages of the devices herein include an inventive metal stack, which by inserting a thin to very thin optically transparent ohmic contact layer, e.g., an n-GaN ohmic contact layer, in between an N-type layer, e.g., n-GaN, and a reflective layer (e.g., an Ag or Au mirror). very good ohmic contact with n-GaN is achieved, while achieving high reflectivity. Such devices are able to achieve low contact resistance (1% work function (Φ) reduction compared to Al baseline), and much higher optical extraction (1.5% light output power (LOP) gain as compared to an Al baseline).

(17) In one or more embodiments, the metal stacks of layers herein are effective as an ohmic contact-reflective material for providing low resistance and high reflectivity in LED devices.

(18) In one or more embodiments, the metal stacks of layers herein are effective as a bonding material for adhering one or more LED devices to another structure, such as substrate, or a circuit board, or a backplane, or the like.

(19) Turning to the figures, FIG. 1 illustrates a cross-sectional view of a metal stack of layers according to one or more embodiments, and FIG. 6 provides an exemplary process flow diagram 550 for manufacture of a metal stack. Methods of depositing materials, layers, and thin films herein include but are not limited to: sputter deposition, evaporation deposition, atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), plasma enhanced atomic layer deposition (PEALD), plasma enhanced chemical vapor deposition (PECVD), and combinations thereof.

(20) Metal stack of layers **100** comprises an ohmic contact layer **151**, which is a layer to be in electrical contact, in particular in one or more embodiments, in direct contact, with an N-layer of a light emitting diode (LED) device. At operation **551** of FIG. **6**, the ohmic contact layer is deposited so that it is electrically contacting the N-type layer of a plurality of semiconductor layers. In one or more embodiments, the ohmic contact layer is in direct contact with the N-type layer. The ohmic contact layer **151** has a work function value that is less than or equal to a work function value of the N-type layer. For example, an N-type layer of n-GaN has a work function value of approximately 4.1 eV. Exemplary ohmic contact layers according to one or more embodiments comprise Al (Vf ~4.06-4.26 eV), Ti (Vf ~4.0-4.33), or AZO (Al-doped ZnO) (Vf of ~3.1-4.5). In one or more embodiments, the ohmic contact layer comprises a thickness in a range of greater than or equal to 5 Å to less than or equal to 200 Å, and all values and subranges therebetween. The ohmic contact layer is designed to be thick enough to form ohmic contact with the N-layer and offer sufficient adhesive force for a reflective layer **153** (e.g., Ag or Au), while not being too thick such that it does not degrade reflectivity of the reflective layer.

(21) Further, it is expected that the ohmic contact layer (e.g., Ti) will react with the N-layer (e.g., n-GaN) at mesa during downstream thermal cycle steps at an interface, which forms, e.g., TiN_x at n-GaN/Ti interface, which further reduce contact resistance and improve the transparency. In FIG. **5**, a transmission electron microscope (TEM) micrograph of an excerpt of an exemplary embodiment made according to methods herein shows a GaN layer on which a Ti ohmic contact layer is deposited and an Ag reflective layer is on the Ti layer. Reaction between the GaN and Ti at their interface is shown by arrow “I”.

(22) The metal stack of layers **100** also comprises the reflective layer **153** electrically contacting the ohmic contact layer **151**. At operation **553** of FIG. **6**, the reflective layer **153** is deposited in direct contact with the ohmic contact layer **151**. In one or more embodiments, the reflective layer **153** comprises silver (Ag) or gold (Au). In one or more embodiments, the reflective layer comprises a thickness of greater than or equal to 1000 Å. In some embodiments, the ohmic contact layer is advantageously very thin relative to the reflective layer (e.g., greater than or equal to 5 Å to less than or equal to 200 Å). In one or more embodiments, the ohmic contact layer has a thickness of less than or equal to 20% a thickness of the reflective layer, including less than or equal to 10%, 5%, 1%, or 0.5%, and all values and subranges therebetween.

(23) The metal stack of layers **100** also comprises a first material barrier layer **157** electrically contacting the reflective layer **153**. At operation **557** of FIG. **6**, the first material barrier layer **157** is deposited in electrical contact with the reflective layer **153**. In one or more embodiments, the first material barrier layer **157** is in direct contact with the reflective layer **153**. In other embodiments, when a first material migration suppression layer **155** is deposited in accordance with optional operation **555** of FIG. **6**, the first material barrier layer **157** is in indirect contact with the reflective layer **153**. The first material barrier layer **157** impedes and/or prevents metals of the reflective layer **153** from forming an intermetallic species or alloy with metals of a current carrying layer **159**. In one or more embodiments, the first material barrier layer **157** comprises: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), tungsten (W), or combinations thereof. In one or more embodiments, the first material barrier layer **157** comprises a thickness of greater than or equal to 1000 Å.

(24) The metal stack of layers **100** also optionally comprises the first material migration suppression layer **155**. In one or more embodiments, the first material migration suppression layer **155** is in direct contact with the reflective layer **153** on one surface and the first material barrier layer **157** on the other surface. The first material migration suppression layer **155** can suppress thermal and electrical migration of metals of the reflective layer **153**. In one or more embodiments, the first material migration suppression layer **155** comprises nickel (Ni) or palladium (Pd). In one or more embodiments, the first material migration suppression layer **155** comprises a thickness of greater than or equal to 50 Å to less than or equal to 1000 Å, and all values and subranges

therebetween.

(25) The metal stack of layers **100** comprises the current carrying layer **159** electrically contacting the first material barrier layer **157**. At operation **559** of FIG. **6**, the current carrying layer **159** is deposited on the first material barrier layer **157**. In one or more embodiments, the current carrying layer **159** is in direct contact with the first material barrier layer **157**. In one or more embodiments, the current carrying layer **159** comprises copper (Cu), gold (Au), aluminum (Al), or combinations thereof. In one or more embodiments, the current carrying layer comprises a thickness of greater than or equal to 5000 Å.

(26) The metal stack of layers **100** comprises a second material barrier layer **163** electrically contacting the current carrying layer **159** and capping the metal stack of layers **100**. At operation **563** of FIG. **6**, the second material barrier layer **163** is deposited in electrical contact with the current carrying layer **159**. In one or more embodiments, the second material barrier layer **163** is in direct contact with the current carrying layer **159**. In other embodiments, when a second material migration suppression layer **161** is deposited in accordance with optional operation **561** of FIG. **6**, the second material barrier layer **163** is in indirect contact with the current carrying layer **159**. The second material barrier layer **163** is a protective layer, which impedes and/or prevents attack of the metal stack by downstream processing steps, i.e. dry etch, wet etch, cleaning etc. In one or more embodiments, the second material barrier layer **163** comprises: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), tungsten (W), or combinations thereof. In one or more embodiments, the second material barrier layer **163** comprises a thickness of greater than or equal to 1000 Å.

(27) The metal stack of layers **100** also optionally comprises the second material migration suppression layer **161**. In one or more embodiments, the second material migration suppression layer **161** is in direct contact with current carrying layer **159** on one surface and the second material barrier layer **163** on the other surface. The second material migration suppression layer **161** can suppress thermal and electrical migration of metals of the current carrying layer **159**. In one or more embodiments, the second material migration suppression layer **161** comprises nickel (Ni) or palladium (Pd). In one or more embodiments, the second material migration suppression layer **161** comprises a thickness of greater than or equal to 50 Å to less than or equal to 1000 Å, and all values and subranges therebetween. In one or more embodiments, the second material barrier layer **163** alone or in combination with the second material migration suppression layer **161** is effective as a capping layer for the metal stack **100** as a whole, to protect against degradation during downstream process steps.

(28) FIG. **7** provides an exemplary process flow diagram **500** for manufacture of a LED device according to one or more general embodiments. At operation **510**, semiconductor layers including N-type layer, active region, and P-type layer are formed according to methods known in the art. In one or more embodiments, the semiconductor layers are formed by epitaxial (EPI) growth. The semiconductor layers according to one or more embodiments comprise epitaxial layers, III-nitride layers, or epitaxial III-nitride layers. In one or more embodiments, the semiconductor layers comprise a III-nitride material, and in specific embodiments epitaxial III-nitride material. In some embodiments, the III-nitride material comprises one or more of gallium (Ga), aluminum (Al), and indium (In). Thus, in some embodiments, the semiconductor layers comprises one or more of gallium nitride (GaN), aluminum nitride (AlN), indium nitride (InN), aluminum gallium nitride (AlGaN), indium gallium nitride (InGaN), indium aluminum nitride (InAlN), aluminum indium gallium nitride (AlInGaN) and the like.

(29) In one or more embodiments, semiconductor layers are grown on a substrate, which may remain or may be removed at later points during manufacturing. The substrate may be any substrate known to one of skill in the art. In one or more embodiments, the substrate comprises one or more of sapphire, silicon carbide, silicon (Si), quartz, magnesium oxide (MgO), zinc oxide (ZnO), spinel, and the like. In one or more embodiments, the substrate is not patterned prior to the growth of the

epitaxial layer(s). Thus, in some embodiments, the substrate is not patterned and can be considered to be flat or substantially flat. In other embodiments, the substrate is patterned, e.g. patterned sapphire substrate (PSS).

(30) In one or more embodiments, the semiconductor layers comprise a stack of undoped III-nitride material and doped III-nitride material. The III-nitride materials may be doped with one or more of silicon (Si), oxygen (O), boron (B), phosphorus (P), germanium (Ge), manganese (Mn), or magnesium (Mg) depending upon whether p-type or n-type III-nitride material is needed. In one or more embodiments, the semiconductor layers have a combined thickness in a range of from about 2 μm to about 10 μm , and all values and subranges therebetween.

(31) In one or more embodiments, the active layer is formed between the N-type layer and the P-type layer. The active layer may comprise any appropriate materials known to one of skill in the art. In one or more embodiments, the active layer is comprised of a III-nitride material multiple quantum wells (MQW), and a III-nitride electron blocking layer.

(32) At operation **515**, dielectric material is deposited. Suitable dielectric materials include, but are not limited to, silicon oxide (SiO), silicon nitride (SiN), silicon carbide (SiC), aluminum oxide (AlOx), aluminum nitride (AlN) and combinations thereof. The skilled artisan will recognize that the use of formulas like SiO, to represent silicon oxide, does not imply any particular stoichiometric relationship between the elements. The formula merely identifies the primary elements of the film.

(33) At operation **520**, etching or patterning is conducted to expose at least the N-type layer. According to one or more embodiments, conventional masking, wet etching and/or dry etching processes can be used.

(34) At operation **550**, a metal stack, suitable as a bonding layer and/or as an ohmic contact-reflective layer is deposited in contact with at least the N-type layer. Operation **550** of forming the metal stack is detailed with respect to FIG. 6. At operations **570** and **575**, any further deposition of materials and/or further etching is conducted to arrive at a desired device configuration.

(35) At operation **580**, any further post-processing is performed. In one or more embodiments, further processing including formation of a passivation layer around a portion or the entirety of a LED or a uLED or the device as a whole. In one or more embodiments, the processed structure retains a substrate, is singulated, and is further processed. In one or more embodiments, the processed structure is flipped and affixed to a support, for example, a tape support, and the substrate is removed. Removal of the substrate is in accordance with methods known in the art including substrate laser liftoff. Upon removal of the substrate, singulated LEDs or uLEDs are created.

(36) Further processing can include deposition of a down-converter material, e.g. layers of a phosphor material.

(37) FIG. 2 illustrates a cross-sectional view of an LED device according to one or more embodiments. In the embodiment of FIG. 2, a chip scale package **200** comprises semiconductor layers including an N-type layer **204n**, an active region **206**, and a P-type layer **204p**. The semiconductor layers reside on a substrate **202**.

(38) An N-bonding material **200n** electrically contacts the N-type layer **204n**. The N-bonding material **200n** is multi-layered, analogous to **100** exemplified in FIG. 1, and comprises an ohmic contact layer **151** electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer; a reflective layer **153** electrically contacting the ohmic contact layer **151**; a material barrier layer **157**, e.g., for FIG. 2, a first N-bonding material barrier layer, electrically contacting the reflective layer; a current carrying layer **159** electrically contacting the material barrier layer **157** (e.g., the first N-bonding material barrier layer); and another material barrier layer **161**, e.g., for FIG. 2, a second N-bonding material barrier layer electrically contacting the current carrying layer **159**. In this embodiment, the ohmic contact layer **151** is in direct contact with the N-type layer **204n**.

- (39) A P-bonding material **200p** electrically contacts the P-type layer **204p** by way of a P-metal layer **212**, a guardsheet **214**, and a current spreading layer **208**. The P-bonding material **200p** is isolated from the N-bonding material **200n** by second dielectric layer **216** and third dielectric layer **218**. In one or more embodiments, the P-bonding material **200p** is the same as the N-bonding material **200n**.
- (40) A first dielectric material **210** insulates the P-type layer **204p** and the active region **206** from the N-bonding material **200n**. The first dielectric material **210** of FIG. 2 also isolates the current spreading layer **208** on the P-type layer **204p** from the N-type layer **204n**.
- (41) The P-metal layer **212** resides on the first dielectric material **210** and in a via opening **207** for direct contact with the current spreading layer **208** and for electrical contact with the P-type layer. The guardsheet **214** is on the p-metal layer **212**.
- (42) An N-pad **230** contacts the N-bonding layer **200n** through an n-opening **219** in the third dielectric layer **218**. The N-pad **230** may be multi-layered, comprising one or more of: a seed layer and various n-electrode metals.
- (43) A P-pad **220** contacts the P-bonding layer **200p** through an p-opening **217** in the third dielectric layer **218**. The P-pad **220** may be multi-layered, comprising one or more of: a seed layer and various p-electrode metals.
- (44) FIG. 3 illustrates a cross-sectional view of an LED device according to one or more embodiments. In the embodiment of FIG. 3, a monolithic die of micro-light emitting diodes (uLEDs) **300** comprises a plurality of mesas **350**, **351** in the form of an array. Mesas **350** and **351** comprises semiconductor layers including an N-type layer **304n**, an active region **306**, and a P-type layer **304p**. In this embodiment, the ohmic contact layer **151** is in direct contact with the N-type layer **304n**.
- (45) A current spreading layer **308** is on the P-type layer **304p**. There is not a substrate in this embodiment. Rather, a first portion **301** of the N-type layer **304n** forms a monolithic body, and a second portion **303** of the N-type layer in combination with the active region, and the P-type layer form a plurality of mesas **350**, **351** integral to the monolithic body. A P-metal layer **312** on the current spreading layer **308** on mesa **350** yields an active pixel and light upon application of current, and in conjunction with an anode and a cathode. A P-metal layer is not included on mesa **351**, which yields an inactive area, namely no light, upon application of current. The mesas **350**, **351** are typically formed by etching a structure of semiconductor layers.
- (46) An N-bonding material **300n** electrically contacts the first portion **301** of the N-type layer **304n** through an N-opening **319** in a dielectric material **316**. The N-bonding material is multi-layered, analogous to FIG. 1, and comprises an ohmic contact layer **151** electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer; a reflective layer **153** electrically contacting the ohmic contact layer **151**; a material barrier layer **157**, e.g., for FIG. 3, a first N-bonding material barrier layer, electrically contacting the reflective layer; a current carrying layer **159** electrically contacting the material barrier layer **157** (e.g., the first N-bonding material barrier layer); and another material barrier layer **161**, e.g., for FIG. 2, a second N-bonding material barrier layer electrically contacting the current carrying layer **159**.
- (47) A P-bonding material **300p** electrically contacts the P-type layer **304p** through a p-opening **317** in the dielectric material **316** by way of a P-metal layer **312** and the current spreading layer **308**. The P-bonding material **300p** is isolated from the N-bonding material **300n** by physically being separate. In one or more embodiments, the P-bonding material **300p** is the same as the N-bonding material **300n**.
- (48) The dielectric layer material **316** insulates the P-type layer and the active region from the N-bonding material. The dielectric material **316** of FIG. 3 also isolates the current spreading layer **308** on the P-type layer **304p** from the N-type layer **304n**.
- (49) FIG. 4A illustrates a cross-sectional view of an LED device according to one or more

embodiments. In the embodiment of FIG. 4, an array of micro-light emitting diodes (uLEDs) **400** comprises a plurality of mesas **450**. Mesas **450** each comprise semiconductor layers including an N-type layer **404n**, an active region **406**, and a P-type layer **404p**. The mesas **450** are typically formed by etching or patterning a structure of semiconductor layers on a substrate such that mesas of the semiconductor layers remain and trenches **452** each having a bottom defined by exposed surfaces of the substrate between the mesas are created, as illustrated in FIG. 4B. In some embodiments, after further construction of the device including filling the trenches **452** with layers and/or materials, the substrate may be removed.

(50) With respect to etching, according to one or more embodiments, highly anisotropic etch methods are used to achieve angles ranging from vertical (90 degrees) to 80 degrees to even smaller values, and all values therebetween. In one or more embodiments, depth of the mesa/trench etch will typically not exceed 5 micrometers. In one or more embodiments, an anisotropic etch is used to form trenches. In one or more embodiments, the width of the mesa is less than 100 micrometers. In one or more embodiments, the height of the mesa is less than or equal to the width of the mesa.

(51) Each mesa has a top surface **453** (usually the top of the P-type layer **404p**) and at least one mesa sidewall **455**. In one or more embodiments, the semiconductor layers are on a substrate **402**. A plurality of trenches **452** between the mesas **450** defined by respective mesa sidewalls **455** and each having a bottom surface **457**, each of the trenches **452** containing an N-ohmic contact-reflective material **400n** electrically contacting the N-type layer **404n** of each of the mesas **450**. The N-ohmic contact-reflective material **400n** is multi-layered, as exemplified in FIG. 1, and comprises an ohmic contact layer **151** electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer; a reflective layer **153** electrically contacting the ohmic contact layer **151**; a material barrier layer **157** electrically contacting the reflective layer; a current carrying layer **159** electrically contacting the material barrier layer **157**; and another material barrier layer **161** electrically contacting the current carrying layer **159**. In this embodiment, the ohmic contact layer **151** is in direct contact with the N-type layer **404n**.

(52) On top surfaces **453** of each mesa **450**, a current spreading layer **408** is on the P-type layer **404p**. Dielectric materials **409**, **410**, and **411** provide electrical isolations. For example, first dielectric material **409** insulates the P-type layer **404p** and the active region **406** from the N-ohmic contact-reflective material **400n**. The first dielectric material **409** also isolates the current spreading layer **408** on the P-type layer **404p** from the N-type layer **404n**.

(53) A P-ohmic contact-reflective material **400p** electrically contacts the P-type layer **404p** through a via **407** in second dielectric material **410** by way of a P-metal layer **412** and the current spreading layer **408**. The P-ohmic contact-reflective material **400p** is isolated from the N-ohmic contact-reflective material **400n** by the first dielectric material **409** and the third dielectric material **411**. In one or more embodiments, the P-ohmic contact-reflective material **400p** is the same as the N-ohmic contact-reflective material **400n**.

(54) The device further comprises electrode metals: N-electrode metal **434** contained by the N-ohmic contact-reflective material **400n**, and P-electrode metal **424** contained by the P-ohmic contact-reflective material **400p**.

(55) In one or more embodiments, a passivation layer **440** is provided on the first dielectric material **409**, and the N-electrode metal **434**. Underbump metallization **442** is optionally on each P-electrode metal **424**, which may be in the form of a plug.

Display Devices

(56) Some display devices comprise single or singulated LEDs or pixels, which include the bonding and/or ohmic contact-reflective material disclosed herein.

(57) Other display devices comprise arrays and groups of LEDs or pixels, which include the bonding and/or ohmic contact-reflective material disclosed herein.

(58) FIG. 8 shows a top plan view of an LED monolithic array **800** comprising a plurality of pixels arranged in a grid of 6×19 . Pixels **855a** and **855b** are examples. In this embodiment, a common cathode **840** is connected to the pixels. Anodes, not shown, present on the underside are included with each pixel. In one or more embodiments, the array comprises an arrangement of 2×2 mesas, 4×4 mesas, 20×20 mesas, 50×50 mesas, 100×100 mesas, or $n1 \times n2$ mesas, where each of $n1$ and $n2$ is a number in a range of from 2 to 1000, and $n1$ and $n2$ can be equal or not equal.

(59) In one or more embodiments, arrays of micro-LEDs (μ LEDs or uLEDs) are used. Micro-LEDs can support high density pixels having a lateral dimension less than $100 \mu\text{m}$ by $100 \mu\text{m}$. In some embodiments, micro-LEDs with dimensions of about $50 \mu\text{m}$ in diameter or width and smaller can be used. Such micro-LEDs can be used for the manufacture of color displays by aligning in close proximity micro-LEDs comprising red, blue and green wavelengths.

(60) In some embodiments, the light emitting arrays include small numbers of micro-LEDs positioned on substrates that are centimeter scale area or greater. In some embodiments, the light emitting arrays include micro-LED pixel arrays with hundreds, thousands, or millions of light emitting LEDs positioned together on centimeter scale area substrates or smaller. In some embodiments, micro-LEDs can include light emitting diodes sized between 30 microns and 500 microns. The light emitting array(s) can be monochromatic, RGB, or other desired chromaticity. In some embodiments, pixels can be square, rectangular, hexagonal, or have curved perimeter. Pixels can be of the same size, of differing sizes, or similarly sized and grouped to present larger effective pixel size.

(61) In some embodiments, light emitting pixels and circuitry supporting light emitting arrays are packaged and optionally include a submount or printed circuit board connected for powering and controlling light production by semiconductor LEDs. In certain embodiments, a printed circuit board supporting light emitting array includes electrical vias, heat sinks, ground planes, electrical traces, and flip chip or other mounting systems. The submount or printed circuit board may be formed of any suitable material, such as ceramic, silicon, aluminum, etc. If the submount material is conductive, an insulating layer is formed over the substrate material, and the metal electrode pattern is formed over the insulating layer. The submount can act as a mechanical support, providing an electrical interface between electrodes on the light emitting array and a power supply, and also provide heat sink functionality.

(62) In some embodiments, LED light emitting arrays include optical elements such as lenses, metalenses, and/or pre-collimators. Optical elements can also or alternatively include apertures, filters, a Fresnel lens, a convex lens, a concave lens, or any other suitable optical element that affects the projected light from the light emitting array. Additionally, one or more of the optical elements can have one or more coatings, including UV blocking or anti-reflective coatings. In some embodiments, optics can be used to correct or minimize two-or three dimensional optical errors including pincushion distortion, barrel distortion, longitudinal chromatic aberration, spherical aberration, chromatic aberration, field curvature, astigmatism, or any other type of optical error. In some embodiments, optical elements can be used to magnify and/or correct images.

Advantageously, in some embodiments magnification of display images allows the light emitting array to be physically smaller, of less weight, and require less power than larger displays.

Additionally, magnification can increase a field of view of the displayed content allowing display presentation equals a user's normal field of view.

Applications

(63) FIG. 9 schematically illustrates an exemplary display system **900** utilizing LEDs disclosed herein. The display system **900** comprises an LED light emitting array **902** and display **908** in electrical communication with an LED driver **904**. The display system **900** also comprises a system controller **906**, such as a microprocessor. The controller **906** is coupled to the LED driver **904**. The controller **906** may also be coupled to the display **908** and to optional sensor(s) **910**, and be powered by power source **912**. In one or more embodiments, user data input is provide to system

controller **906**.

(64) In one or more embodiments, the system is a camera flash system utilizing uLEDs. In such an embodiment, the LED light emitting array **902** is an illumination array and lens system and the display **908** comprises a camera, wherein the LEDs of **902** and the camera of **908** may be controlled by the controller **906** to match their fields of view.

(65) Optionally sensors **910** with control input may include, for example, positional sensors (e.g., a gyroscope and/or accelerometer) and/or other sensors that may be used to determine the position, speed, and orientation of system. The signals from the sensors **910** may be supplied to the controller **906** to be used to determine the appropriate course of action of the controller **906** (e.g., which LEDs are currently illuminating a target and which LEDs will be illuminating the target a predetermined amount of time later).

(66) In operation, illumination from some or all of the pixels of the LED array in **902** may be adjusted—deactivated, operated at full intensity, or operated at an intermediate intensity. As noted above, beam focus or steering of light emitted by the LED array in **902** can be performed electronically by activating one or more subsets of the pixels, to permit dynamic adjustment of the beam shape without moving optics or changing the focus of the lens in the lighting apparatus.

(67) LED array systems such as described herein may support various other beam steering or other applications that benefit from fine-grained intensity, spatial, and temporal control of light distribution. These applications may include, but are not limited to, precise spatial patterning of emitted light from pixel blocks or individual pixels. Depending on the application, emitted light may be spectrally distinct, adaptive over time, and/or environmentally responsive. The light emitting pixel arrays may provide pre-programmed light distribution in various intensity, spatial, or temporal patterns. Associated optics may be distinct at a pixel, pixel block, or device level. An example light emitting pixel array may include a device having a commonly controlled central block of high intensity pixels with an associated common optic, whereas edge pixels may have individual optics. In addition to flashlights, common applications supported by light emitting pixel arrays include video lighting, automotive headlights, architectural and area illumination, and street lighting.

(68) Other applications of LED devices herein include an augmented reality/virtual reality (AR/VR) systems, which may utilize uLEDs disclosed herein. One or more AR/VR systems include: augmented (AR) or virtual reality (VR) headsets, glasses, or projectors. Such AR/VR systems includes an LED light emitting array, an LED driver (or light emitting array controller), a system controller, an AR or VR display, a sensor system **810**. Control input may be provided to the sensor system, while power and user data input is provided to the system controller. As will be understood, in some embodiments modules included in the AR/VR system can be compactly arranged in a single structure, or one or more elements can be separately mounted and connected via wireless or wired communication. For example, the light emitting array, AR or VR display, and sensor system can be mounted on a headset or glasses, with the LED driver and/or system controller separately mounted.

(69) In one embodiment, the light emitting array can be used to project light in graphical or object patterns that can support AR/VR systems. In some embodiments, separate light emitting arrays can be used to provide display images, with AR features being provided by a distinct and separate micro-LED array. In some embodiments, a selected group of pixels can be used for displaying content to the user while tracking pixels can be used for providing tracking light used in eye tracking. Content display pixels are designed to emit visible light, with at least some portion of the visible band (approximately 400 nm to 750 nm). In contrast, tracking pixels can emit light in visible band or in the IR band (approximately 750 nm to 2,200 nm), or some combination thereof. As an alternative example, the tracking pixels could operate in the 800 to 1000 nanometer range. In some embodiments, the tracking pixels can emit tracking light during a time period that content pixels are turned off and are not displaying content to the user.

(70) The AR/VR system can incorporate a wide range of optics in the LED light emitting array and/or AR/VR display, for example to couple light emitted by the LED light emitting array into AR/VR display as discussed above. For AR/VR applications, these optics may comprise nanofins and be designed to polarize the light they transmit.

(71) In one embodiment, the light emitting array controller can be used to provide power and real time control for the light emitting array. For example, the light emitting array controller can be able to implement pixel or group pixel level control of amplitude and duty cycle. In some embodiments, the light emitting array controller further includes a frame buffer for holding generated or processed images that can be supplied to the light emitting array. Other supported modules can include digital control interfaces such as Inter-Integrated Circuit (I2C) serial bus, Serial Peripheral Interface (SPI), USB-C, HDMI, Display Port, or other suitable image or control modules that are configured to transmit needed image data, control data or instructions.

(72) In operation, pixels in the images can be used to define response of corresponding light emitting array, with intensity and spatial modulation of LED pixels being based on the image(s). To reduce data rate issues, groups of pixels (e.g. 5×5 blocks) can be controlled as single blocks in some embodiments. In some embodiments, high speed and high data rate operation is supported, with pixel values from successive images able to be loaded as successive frames in an image sequence at a rate between 30 Hz and 100 Hz, with 60 Hz being typical. Pulse width modulation can be used to control each pixel to emit light in a pattern and with an intensity at least partially dependent on the image.

(73) In some embodiments, the sensor system can include external sensors such as cameras, depth sensors, or audio sensors that monitor the environment, and internal sensors such as accelerometers or two or three axis gyroscopes that monitor AR/VR headset position. Other sensors can include but are not limited to air pressure, stress sensors, temperature sensors, or any other suitable sensors needed for local or remote environmental monitoring. In some embodiments, control input can include detected touch or taps, gestural input, or control based on headset or display position. As another example, based on the one or more measurement signals from one or more gyroscope or position sensors that measure translation or rotational movement, an estimated position of AR/VR system relative to an initial position can be determined.

(74) In some embodiments, the system controller uses data from the sensor system to integrate measurement signals received from the accelerometers over time to estimate a velocity vector and integrate the velocity vector over time to determine an estimated position of a reference point for the AR/VR system. In other embodiments, the reference point used to describe the position of the AR/VR system can be based on depth sensor, camera positioning views, or optical field flow.

(75) Based on changes in position, orientation, or movement of the AR/VR system, the system controller can send images or instructions the light emitting array controller. Changes or modification the images or instructions can also be made by user data input, or automated data input as needed. User data input can include but is not limited to that provided by audio instructions, haptic feedback, eye or pupil positioning, or connected keyboard, mouse, or game controller.

EMBODIMENTS

(76) Various embodiments are listed below. It will be understood that the embodiments listed below may be combined with all aspects and other embodiments in accordance with the scope of the invention.

(77) Embodiment (a). A light emitting diode (LED) device comprising: a plurality of mesas, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each mesa having a top surface and at least one mesa sidewall; a plurality of trenches between the mesas defined by respective mesa sidewalls and each having a bottom surface, each of the trenches containing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas, the N-ohmic contact-reflective

material being multi-layered, and comprising: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, a reflective layer electrically contacting the ohmic contact layer, a first material barrier layer electrically contacting the reflective layer, a current carrying layer electrically contacting the first material barrier layer, and a second material barrier layer electrically contacting the current carrying layer; an N-electrode metal contained by the N-ohmic contact-reflective material, a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material; and a P-electrode metal in electrical contact with the P-type layer of each of the mesas.

(78) Embodiment (b). The LED device of embodiment (a), wherein the N-ohmic contact-reflective material is in direct contact with the N-type layer.

(79) Embodiment (c). The LED device of any of embodiments (a) to (b), wherein the N-type layer comprises n-GaN and: the ohmic contact layer comprises: aluminum (Al), titanium (Ti), or aluminum-doped zinc oxide (AZO), the reflective layer comprises silver (Ag) or gold (Au); the first and second material barrier layers each independently comprise: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), or tungsten (W), and the current carrying layer comprises: copper (Cu), gold (Au), or aluminum (Al).

(80) Embodiment (d). The LED device of any of embodiments (a) to (c), wherein: the ohmic contact layer comprises a thickness in a range of greater than or equal to 5 Å to less than or equal to 200 Å, the reflective layer comprises a thickness of greater than or equal to 1000 Å, the first and second material barrier layers each independently comprise a thickness of greater than or equal to 1000 Å, and the current carrying layer comprises a thickness of greater than or equal to 5000 Å.

(81) Embodiment (e). The LED device of any of embodiments (a) to (d), wherein the N-ohmic contact-reflective material further comprises a first material migration suppression layer electrically contacting the reflective layer and the first material barrier layer; and a second material migration suppression layer electrically contacting the current carrying layer and the second material barrier layer.

(82) Embodiment (f). The LED device of embodiment (e), wherein the first and second material migration suppression layers each independently comprise nickel (Ni) or palladium (Pd), and/or independently comprise a thickness in a range of greater than or equal to 50 Å to less than or equal to 1000 Å.

(83) Embodiment (g). The LED device of any of embodiments (a) to (f) further comprising a P-ohmic contact-reflective material containing the P-electrode metal, wherein the P-ohmic contact-reflective material has the same structure as the N-ohmic contact-reflective material.

(84) Embodiment (h). The LED device of any of embodiments (a) to (g), wherein each of the mesas has at least one characteristic dimension of less than 100 micrometers, the character dimension being selected from the group consisting of: height, width, and depth.

(85) Embodiment (i). The LED device of any of embodiments (a) to (h), wherein the semiconductor layers are epitaxial semiconductor layers having an overall thickness in a range of from 2 μm to 10 μm.

(86) Embodiment (j). The LED device of any of embodiments (a) to (i), wherein each of the mesas includes sidewalls of the semiconductor layers defining an angle in a range of from 60 degrees to 90 degrees from a horizontal plane that is parallel with the N-type layer and the P-type layer.

(87) Embodiment (k). The LED device of any of embodiments (a) to (j), wherein the plurality of mesas are integral to a monolithic die.

(88) Embodiment (l). The LED device of any of embodiments (a) to (j), wherein the semiconductor layers are on a substrate.

(89) Embodiment (m). A method of manufacturing a light emitting diode (LED) device comprising: preparing a plurality of mesas and trenches, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each

mesa having a top surface and at least one mesa sidewall, and the trenches defined by respective mesa sidewalls and each having a bottom surface; exposing the N-type layer; preparing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas by: depositing an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, depositing a reflective layer electrically contacting the ohmic contact layer, depositing a first material barrier layer electrically contacting the reflective layer, depositing a current carrying layer electrically contacting the first material barrier layer, and depositing a second material barrier layer electrically contacting the current carrying layer; depositing and patterning an N-electrode metal contained by the N-ohmic contact-reflective material and a P-electrode metal in electrical contact with the P-type layer of each of the mesas; and depositing and patterning a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material.

(90) Embodiment (n). The method of embodiment (m) comprising depositing the N-ohmic contact-reflective material directly on the N-type layer.

(91) Embodiment (o). The method of any of embodiments (m) to (n), wherein the N-type layer comprises n-GaN, and the ohmic contact layer comprises: aluminum (Al), titanium (Ti), or aluminum-doped zinc oxide (AZO); the reflective layer comprises silver (Ag) or gold (Au); the first and second material barrier layers each independently comprise: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), or tungsten (W); and the current carrying layer comprises: copper (Cu), gold (Au), or aluminum (Al).

(92) Embodiment (p). The method of any of embodiments (m) to (o), wherein: the ohmic contact layer comprises a thickness in a range of greater than or equal to 5 Å to less than or equal to 200 Å, the reflective layer comprises a thickness of greater than or equal to 1000 Å, the first and second material barrier layers each independently comprise a thickness of greater than or equal to 1000 Å, and the current carrying layer comprises a thickness of greater than or equal to 5000 Å.

(93) Embodiment (q). The method of any of embodiments (m) to (p), further comprising: depositing a first material migration suppression layer electrically contacting the reflective layer and the first material barrier layer; and/or a second material migration suppression layer electrically contacting the current carrying layer and the second material barrier layer.

(94) Embodiment (r). The method of any of embodiment (q), wherein the first and second material migration suppression layers each independently comprise nickel (Ni) or palladium (Pd), and/or independently comprise a thickness in a range of greater than or equal to 50 Å to less than or equal to 1000 Å.

(95) Embodiment (s). The method of any of embodiments (m) to (r), wherein a P-bonding material is also prepared contacting the P-type layer by depositing the ohmic contact layer electrically contacting the P-type layer, depositing the reflective layer electrically contacting the ohmic contact layer, depositing the first N-bonding material barrier layer electrically contacting the reflective layer, depositing the current carrying layer electrically contacting the first material barrier layer.

(96) The method of any of embodiments (m) to (s), wherein each of the mesas has at least one characteristic dimension of less than 100 micrometers, the character dimension being selected from the group consisting of: height, width, and depth.

(97) Reference throughout this specification to “one embodiment,” “certain embodiments,” “one or more embodiments” or “an embodiment” means that a particular feature, structure, material, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, the appearances of the phrases such as “in one or more embodiments,” “in certain embodiments,” “in one embodiment” or “in an embodiment” in various places throughout this specification are not necessarily referring to the same embodiment of the disclosure.

Furthermore, the particular features, structures, materials, or characteristics may be combined in any suitable manner in one or more embodiments.

(98) Many modifications and other embodiments of the invention will come to the mind of one

skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the invention is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims. It is also understood that other embodiments of this invention may be practiced in the absence of an element/step not specifically disclosed herein.

Claims

1. A light emitting diode (LED) device comprising: a plurality of mesas, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each mesa having a top surface and at least one mesa sidewall; a plurality of trenches between the mesas defined by respective mesa sidewalls and each having a bottom surface, each of the trenches containing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas, the N-ohmic contact-reflective material being multi-layered, and comprising: an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, a reflective layer electrically contacting the ohmic contact layer, a first material barrier layer electrically contacting the reflective layer, a current carrying layer electrically contacting the first material barrier layer, and a second material barrier layer electrically contacting the current carrying layer; an N-electrode metal contained by the N-ohmic contact-reflective material, a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material; and a P-electrode metal in electrical contact with the P-type layer of each of the mesas.
2. The LED device of claim 1, wherein the N-ohmic contact-reflective material is in direct contact with the N-type layer.
3. The LED device of claim 1, wherein the N-type layer comprises n-GaN and: the ohmic contact layer comprises: aluminum (Al), titanium (Ti), or aluminum-doped zinc oxide (AZO), the reflective layer comprises silver (Ag) or gold (Au); the first and second material barrier layers each independently comprise: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), or tungsten (W), and the current carrying layer comprises: copper (Cu), gold (Au), or aluminum (Al).
4. The LED device of claim 1, wherein: the ohmic contact layer comprises a thickness in a range of greater than or equal to 5 Å to less than or equal to 200 Å, the reflective layer comprises a thickness of greater than or equal to 1000 Å, the first and second material barrier layers each independently comprise a thickness of greater than or equal to 1000 Å, and the current carrying layer comprises a thickness of greater than or equal to 5000 Å.
5. The LED device of claim 1, wherein the N-ohmic contact-reflective material further comprises a first material migration suppression layer electrically contacting the reflective layer and the first material barrier layer; and a second material migration suppression layer electrically contacting the current carrying layer and the second material barrier layer.
6. The LED device of claim 5, wherein the first and second material migration suppression layers each independently comprise nickel (Ni) or palladium (Pd), and/or independently comprise a thickness in a range of greater than or equal to 50 Å to less than or equal to 1000 Å.
7. The LED device of claim 1 further comprising a P-ohmic contact-reflective material containing the P-electrode metal, wherein the P-ohmic contact-reflective material has the same structure as the N-ohmic contact-reflective material.
8. The LED device of claim 1, wherein each of the mesas has at least one characteristic dimension of less than 100 micrometers, the character dimension being selected from the group consisting of: height, width, and depth.
9. The LED device of claim 1, wherein the semiconductor layers are epitaxial semiconductor layers

having an overall thickness in a range of from 2 μm to 10 μm .

10. The LED device of claim 1, wherein each of the mesas includes sidewalls of the semiconductor layers defining an angle in a range of from 60 degrees to 90 degrees from a horizontal plane that is parallel with the N-type layer and the P-type layer.

11. The LED device of claim 1, wherein the plurality of mesas are integral to a monolithic die.

12. The LED device of claim 1, wherein the semiconductor layers are on a substrate.

13. A method of manufacturing a light emitting diode (LED) device comprising: preparing a plurality of mesas and trenches, each of the mesas comprising semiconductor layers, the semiconductor layers including an N-type layer, an active region, and a P-type layer, and each mesa having a top surface and at least one mesa sidewall, and the trenches defined by respective mesa sidewalls and each having a bottom surface; exposing the N-type layer; preparing an N-ohmic contact-reflective material electrically contacting the N-type layer of each of the mesas by: depositing an ohmic contact layer electrically contacting the N-type layer and having a work function value that is less than or equal to a work function value of the N-type layer, depositing a reflective layer electrically contacting the ohmic contact layer, depositing a first material barrier layer electrically contacting the reflective layer, depositing a current carrying layer electrically contacting the first material barrier layer, and depositing a second material barrier layer electrically contacting the current carrying layer; depositing and patterning an N-electrode metal contained by the N-ohmic contact-reflective material and a P-electrode metal in electrical contact with the P-type layer of each of the mesas; and depositing and patterning a dielectric material which insulates the P-type layer and the active region from the N-ohmic contact-reflective material.

14. The method of claim 13 comprising depositing the N-ohmic contact-reflective material directly on the N-type layer.

15. The method of claim 13, wherein the N-type layer comprises n-GaN, and the ohmic contact layer comprises: aluminum (Al), titanium (Ti), or aluminum-doped zinc oxide (AZO); the reflective layer comprises silver (Ag) or gold (Au); the first and second material barrier layers each independently comprise: titanium (Ti), chromium (Cr), platinum (Pt), cobalt (Co), palladium (Pd), or tungsten (W); and the current carrying layer comprises: copper (Cu), gold (Au), or aluminum (Al).

16. The method of claim 13, wherein: the ohmic contact layer comprises a thickness in a range of greater than or equal to 5 $\text{\AA}$ to less than or equal to 200 $\text{\AA}$, the reflective layer comprises a thickness of greater than or equal to 1000 $\text{\AA}$, the first and second material barrier layers each independently comprise a thickness of greater than or equal to 1000 $\text{\AA}$, and the current carrying layer comprises a thickness of greater than or equal to 5000 $\text{\AA}$.

17. The method of claim 13 further comprising: depositing a first material migration suppression layer electrically contacting the reflective layer and the first material barrier layer; and/or a second material migration suppression layer electrically contacting the current carrying layer and the second material barrier layer.

18. The method of claim 17, wherein the first and second material migration suppression layers each independently comprise nickel (Ni) or palladium (Pd), and/or independently comprise a thickness in a range of greater than or equal to 50 $\text{\AA}$ to less than or equal to 1000 $\text{\AA}$.

19. The method of claim 13, wherein a P-bonding material is also prepared contacting the P-type layer by depositing the ohmic contact layer electrically contacting the P-type layer, depositing the reflective layer electrically contacting the ohmic contact layer, depositing the first N-bonding material barrier layer electrically contacting the reflective layer, depositing the current carrying layer electrically contacting the first material barrier layer.

20. The method of claim 13, wherein each of the mesas has at least one characteristic dimension of less than 100 micrometers, the character dimension being selected from the group consisting of: height, width, and depth.
